

Basic probability

- *Conditional Probability:* $\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$
- *Law of Total Probability:* $\{H_1, \dots, H_M\}$ disjoint events such that $H_1 \cup \dots \cup H_M = \Omega \Rightarrow \mathbb{P}(E) = \sum_{m=1}^M \mathbb{P}(E \mid H_m) \mathbb{P}(H_m)$
- *Bayes' Rule:* $\mathbb{P}(H_i \mid E) = \frac{\mathbb{P}(E \mid H_i) \mathbb{P}(H_i)}{\mathbb{P}(E)} \quad \forall i \in \{1, \dots, M\}$
- *Independence:* A and B independent $\Leftrightarrow \mathbb{P}(A \cap B) = \mathbb{P}(A) \mathbb{P}(B)$ or $\mathbb{P}(A \mid B) = \mathbb{P}(A)$
- *Conditional Independence:* A and B are independent given $C \Leftrightarrow \mathbb{P}(A \cap B \mid C) = \mathbb{P}(A \mid C) \mathbb{P}(B \mid C)$ or $\mathbb{P}(A \mid B \cap C) = \mathbb{P}(A \mid C)$

Random variables

Discrete r.v.

- $p_X(x) = \mathbb{P}(X = x) \quad \text{and} \quad F_X(x) = \mathbb{P}(X \leq x) = \sum_{x_i \leq x} p_X(x_i)$
- $p_X(x) = F_X(x) - F_X(x^-)$

Quantile function at level $p \in (0, 1)$

- $q_p = \left\{ \text{The smallest value } x \text{ s.t. } F_X(x) = \mathbb{P}(X \leq x) \text{ is at least } p \right\}$

Continuous r.v.

- $f_X(x) = F'_X(x) \quad \text{and} \quad F_X(x) = \int_{-\infty}^x f_X(t) dt$
- $\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x) dx = F_X(b) - F_X(a)$

Random vectors

Discrete r.v.

- $p_{X,Y}(x, y) = \mathbb{P}(X = x, Y = y)$
- $F_{X,Y}(x, y) = \mathbb{P}(X \leq x, Y \leq y) = \sum_{x_i \leq x} \sum_{y_j \leq y} p_{X,Y}(x_i, y_j)$
- $p_X(x) = \sum_y p_{X,Y}(x, y) \quad \text{and} \quad p_Y(y) = \sum_x p_{X,Y}(x, y)$
- $p_{X|Y}(x|y) = \frac{p_{X,Y}(x,y)}{p_Y(y)} \quad \text{and} \quad p_{Y|X}(y|x) = \frac{p_{X,Y}(x,y)}{p_X(x)}$

Continuous r.v.

- $f_{X,Y}(x, y) = \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x, y) \quad \text{and} \quad F_{X,Y}(x, y) = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(r, t) dr dt$
- $\mathbb{P}(a_1 \leq X \leq b_1, a_2 \leq Y \leq b_2) = \int_{a_1}^{b_1} \int_{a_2}^{b_2} f_{X,Y}(x, y) dx dy$
- $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy \quad \text{and} \quad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx$
- $f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} \quad \text{and} \quad f_{Y|X}(y|x) = \frac{f_{X,Y}(x,y)}{f_X(x)}$

Marginal CDF's

- $F_X(x) = \lim_{y \rightarrow \infty} F_{X,Y}(x, y) \quad \text{and} \quad F_Y(y) = \lim_{x \rightarrow \infty} F_{X,Y}(x, y)$

Independence

- X and Y independent $\Leftrightarrow F_{X,Y}(x, y) = F_X(x) \cdot F_Y(y) \quad \forall (x, y) \in \mathbb{R}^2 \Leftrightarrow f_{X,Y}(x, y) = f_X(x) \cdot f_Y(y) \quad \forall (x, y) \in \mathbb{R}^2$.
- If $\{X_1, \dots, X_n\}$ are independent, then $\{Y_1, \dots, Y_n\}$ with $Y_i = h_i(X_i)$ are independent.
- X and Y are conditional independent given $Z \Leftrightarrow f(x, y|z) = f(x|z)f(y|z) \quad \forall (x, y, z) \in \mathbb{R}^3 \Leftrightarrow f(x|y, z) = f(x|z) \quad \forall (x, y, z) \in \mathbb{R}^3$.

Expectation

Discrete r.v.

- $\mathbb{E}(X) = \sum_i x_i p_X(x_i)$
- $\mathbb{E}(g(X)) = \sum_i g(x_i) p_X(x_i)$
- $\mathbb{E}(X|Y = y_s) = \sum_i x_i \mathbb{P}(X = x_i | Y = y_s)$

Law of Total Expectation

- $\mathbb{E}(X) = \mathbb{E}_Y(\mathbb{E}(X|Y)) \quad \text{and} \quad \mathbb{E}(Y) = \mathbb{E}_X(\mathbb{E}(Y|X))$

Continuous r.v.

- $\mathbb{E}(X) = \int_{-\infty}^{\infty} x f_X(x) dx$
- $\mathbb{E}(g(X)) = \int_{-\infty}^{\infty} g(x) f_X(x) dx$
- $\mathbb{E}(X|Y = y) = \int_{-\infty}^{\infty} x f_{X|Y}(x|y) dx = \int_{-\infty}^{\infty} x \frac{f_{X,Y}(x,y)}{f_Y(y)} dx$

Jensen inequality

- $\mathbb{E}(g(X)) \geq g(\mathbb{E}(X)), \quad g(\cdot) \text{ convex.}$

Variance, Covariance and Correlation

- $\mathbb{V}ar(X) = \mathbb{E}(X - \mathbb{E}(X))^2 = \mathbb{E}(X^2) - (\mathbb{E}(X))^2$
- $\mathbb{V}ar(X|Y = y) = \int_{-\infty}^{\infty} (x - \mathbb{E}(X|Y = y))^2 f_{X|Y}(x|y) dx$
- *Law of Total Variance:* $\mathbb{V}ar(X) = \mathbb{E}_Y(\mathbb{V}ar(X|Y)) + \mathbb{V}ar_Y(\mathbb{E}(X|Y))$
- $\mathbb{C}ov(X, Y) = \mathbb{E}(X - \mathbb{E}(X))(Y - \mathbb{E}(Y)) = \mathbb{E}(X \cdot Y) - \mathbb{E}(X) \cdot \mathbb{E}(Y)$
- $\mathbb{V}ar\left(\sum_i \alpha_i X_i\right) = \sum_i \alpha_i^2 \mathbb{V}ar(X_i) + 2 \sum_{i < j} \alpha_i \alpha_j \mathbb{C}ov(X_i, X_j)$
- $\rho(X, Y) = \rho_{X,Y} = \frac{\mathbb{C}ov(X,Y)}{\sqrt{\mathbb{V}ar(X) \mathbb{V}ar(Y)}} \in [-1, +1]$

“Other” Correlations

- *Partial Correlation:* $\rho_{X,Y|Z} = \frac{\rho_{X,Y} - (\rho_{X,Z} \cdot \rho_{Y,Z})}{\sqrt{1 - \rho_{X,Z}^2} \cdot \sqrt{1 - \rho_{Y,Z}^2}}$ (single confounder); $\mathbf{R}_{jk}^{(P)} = -\frac{\Lambda_{jk}}{\sqrt{\Lambda_{jj}} \cdot \sqrt{\Lambda_{kk}}}$, with $\Lambda = \Sigma^{-1}$ the precision matrix (general)
- *Kendall τ :* $\tau(X, Y) = \mathbb{E}[\text{sign}((X - X')(Y - Y'))]$
- *Spearman ρ :* $\rho^{(S)}(X, Y) = \rho(Z_X, Z_Y)$ with $Z_X = F_X(X)$ and $Z_Y = F_Y(Y)$
- *Distance correlation:* $\rho^2(\mathbf{X}, \mathbf{Y}) = \frac{\gamma^2(\mathbf{X}, \mathbf{Y})}{\sqrt{\gamma^2(\mathbf{X}, \mathbf{X})} \cdot \sqrt{\gamma^2(\mathbf{Y}, \mathbf{Y})}}$ with $\gamma^2(\mathbf{X}, \mathbf{Y}) = \mathbb{E}[\delta(\mathbf{X}, \mathbf{X}') \cdot \delta(\mathbf{Y}, \mathbf{Y}')] \text{ where, if } d(\mathbf{x}, \mathbf{x}') = \|\mathbf{x} - \mathbf{x}'\|,$

$$\delta(\mathbf{X}, \mathbf{X}') = d(\mathbf{X}, \mathbf{X}') - 2 \cdot \int d(\mathbf{X}, \mathbf{u}) \, dF_X(\mathbf{u}) + \int \int d(\mathbf{u}, \mathbf{v}) \, dF_X(\mathbf{u}) \, dF_X(\mathbf{v})$$

Transformations

Change of units: $Y = rX + s$

$$\square F_Y(y) = F_X\left(\frac{y-s}{r}\right) \quad \text{and} \quad f_Y(y) = \frac{1}{r} f_X\left(\frac{y-s}{r}\right)$$

$$\square \mathbb{E}(rX + s) = r \mathbb{E}(X) + s$$

$$\square \mathbb{V}\text{ar}(rX + s) = r^2 \mathbb{V}\text{ar}(X)$$

Transformations: $Y = g(X)$

$$\square p_Y(y) = \mathbb{P}(g(X) = y) = \mathbb{P}(\{x : g(x) = y\}) = \sum_{x \in g^{-1}(y)} p_X(x)$$

$$\square F_Y(y) = \mathbb{P}(g(X) \leq y) = \int_{A_y} f_X(x) \, dx \quad \text{with} \quad A_y = \{x : g(x) \leq y\}$$

$$\square f_Y(y) = F'_Y(y) = f_X(s(y)) |s'(y)| \quad \text{if } g(\cdot) \text{ strictly monotone with inverse } s(\cdot).$$

Quantile and Probability Transforms

□ *Quantile Transform:* $U \sim \text{Unif}(0, 1)$ and $F(\cdot)$ CDF with quantile function $\mathbb{Q}(\cdot)$. $Y = \mathbb{Q}(U)$ has CDF equal to $F(\cdot)$.

□ *Probability Transform:* Y has continuous CDF $F_Y(\cdot)$, then $U = F_Y(Y) \sim \text{Unif}(0, 1)$.

Very Popular Distributions

Discrete models

□ **Bernoulli:** $X \sim \text{Ber}(p)$

- $p_X(x) = p^x(1-p)^{1-x}$, $x \in \{0, 1\}$ and $p \in [0, 1]$
- $\mathbb{E}(X) = p$ and $\mathbb{V}\text{ar}(X) = p(1-p)$
- $\{X_i\}_{i=1}^n \stackrel{\text{i.i.d.}}{\sim} \text{Ber}(p) \Rightarrow \sum_{i=1}^n X_i \sim \text{Binom}(n, p)$

□ **Binomial:** $X \sim \text{Binom}(n, p)$

- $p_X(x) = \binom{n}{x} p^x(1-p)^{n-x}$, $x \in \{0, 1, \dots, n\}$ and $p \in [0, 1]$
- $\mathbb{E}(X) = np$ and $\mathbb{V}\text{ar}(X) = np(1-p)$
- $\{X_i\}_{i=1}^n \stackrel{\text{i.i.d.}}{\sim} \text{Binom}(n_i, p) \Rightarrow \sum_{i=1}^n X_i \sim \text{Binom}(\sum_i n_i, p)$

□ **Geometric:** $X \sim \text{Geo}(p)$

- $p_X(x) = p(1-p)^{x-1}$, $x \in \{1, 2, 3, \dots\}$ and $p \in [0, 1]$
- $\mathbb{E}(X) = \frac{1}{p}$
- $\mathbb{V}\text{ar}(X) = \frac{1-p}{p^2}$

□ **Poisson:** $X \sim \text{Pois}(\mu)$

- $p_X(x) = \frac{\mu^x}{x!} e^{-\mu}$, $x \in \{0, 1, 2, 3, \dots\}$ and $\mu \in \mathbb{R}^+$
- $\mathbb{E}(X) = \mu$ and $\mathbb{V}\text{ar}(X) = \mu$
- $\{X_i\}_{i=1}^n \stackrel{\text{i.i.d.}}{\sim} \text{Pois}(\mu_i) \Rightarrow \sum_{i=1}^n X_i \sim \text{Pois}(\sum_i \mu_i)$

Continuous models

□ **Uniform:** $X \sim \text{Unif}(a, b)$

- $f_X(x) = \frac{1}{b-a} \mathbb{I}_{[a,b]}(x)$, $a < b \in \mathbb{R}$
- $F_X(x) = \frac{x-a}{b-a}$, $x \in [a, b]$
- $\mathbb{E}(X) = \frac{a+b}{2}$ and $\mathbb{V}\text{ar}(X) = \frac{(b-a)^2}{12}$

□ **Exponential:** $X \sim \text{Exp}(\lambda)$

- $f_X(x) = \lambda e^{-\lambda x} \mathbb{I}_{\mathbb{R}_0^+}(x)$, $\lambda > 0$
- $F_X(x) = 1 - e^{-\lambda x}$, $x \geq 0$
- $\mathbb{E}(X) = \frac{1}{\lambda}$ and $\mathbb{V}\text{ar}(X) = \frac{1}{\lambda^2}$

□ **Normal:** $X \sim \text{N}(\mu, \sigma^2)$

- $f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$, $x \in \mathbb{R}$
- $F_X(x) = \int_{-\infty}^x f_X(t) \, dt$, $x \in \mathbb{R}$
- $\mathbb{E}(X) = \mu$ and $\mathbb{V}\text{ar}(X) = \sigma^2$
- $Z = \frac{X-\mu}{\sigma} \sim \text{N}(0, 1)$, and $X = \mu + \sigma Z \sim \text{N}(\mu, \sigma^2)$
- $q_p = \mu + \sigma z_p$
- $\{X_i\}_{i=1}^n \stackrel{\text{i.i.d.}}{\sim} \text{N}(\mu_i, \sigma_i^2) \Rightarrow \sum_{i=1}^n X_i \sim \text{N}(\sum_i \mu_i, \sum_i \sigma_i^2)$

Moment Generating Function

$$\square \psi_X(t) = \mathbb{E}(e^{X \cdot t}) \Rightarrow \mathbb{E}(X^k) = \psi_X^{(k)}(0)$$

$$\square X \perp Y \Leftrightarrow \psi_{X+Y}(t) = \psi_X(t) \cdot \psi_Y(t)$$

Bivariate Normal

□ Let's assume the random vector $[X_1, X_2]^T$ is distributed as a $\text{N}_2(\boldsymbol{\mu}, \Sigma)$, with $\boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$ and $\Sigma = \begin{bmatrix} \sigma_1^2 & \rho \sigma_1 \sigma_2 \\ \rho \sigma_1 \sigma_2 & \sigma_2^2 \end{bmatrix}$, then

$$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)} \left[\frac{(x_1-\mu_1)^2}{\sigma_1^2} + \frac{(x_2-\mu_2)^2}{\sigma_2^2} - \frac{2\rho(x_1-\mu_1)(x_2-\mu_2)}{\sigma_1\sigma_2} \right]\right)$$

□ The conditional distribution: $(X_1 | X_2 = x_2) \sim \text{N}_1\left(\mu_1 + \frac{\sigma_1}{\sigma_2} \rho (x_2 - \mu_2), (1 - \rho^2) \sigma_1^2\right) \Rightarrow \begin{cases} \mathbb{E}(X_1 | X_2 = x_2) = \mu_1 + \frac{\sigma_1}{\sigma_2} \rho (x_2 - \mu_2) \\ \mathbb{V}\text{ar}(X_1 | X_2 = x_2) = (1 - \rho^2) \sigma_1^2 \end{cases}$

Inequalities

□ **Chebyshev:** $\mathbb{P}\left(|\bar{X}_n - \mu| < t\right) \geq 1 - \frac{\sigma^2}{n t^2}$

□ $C_\alpha^{(c)} = \left[\bar{X}_n - t_\alpha^{(c)}, \bar{X}_n + t_\alpha^{(c)}\right]$ with $t_\alpha^{(c)} = \frac{\sigma}{\sqrt{n\alpha}}$

□ **Hoeffding:** $\mathbb{P}\left(|\bar{X}_n - \mu| < t\right) \geq 1 - 2 \cdot e^{-\frac{2 n t^2}{(b-a)^2}}, \quad \text{if } X_i \in [a, b]$

□ $C_\alpha^{(c)} = \left[\bar{X}_n - t_\alpha^{(H)}, \bar{X}_n + t_\alpha^{(H)}\right]$ with $t_\alpha^{(H)} = \sqrt{\frac{(b-a)^2}{2n} \log\left(\frac{2}{\alpha}\right)}$

Convergence

□ **WLLN:** $\bar{X}_n \xrightarrow{p} \mu$, with $\mu = \mathbb{E}(X)$

□ **CLT:** $\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \rightsquigarrow Z \sim N(0, 1)$ or $\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \dot{\sim} N(0, 1)$

□ **“Studentized”:** $\frac{\sqrt{n}(\bar{X}_n - \mu)}{S_n} \rightsquigarrow N(0, 1)$, $S_n = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$

□ **δ -method:** $X_n \dot{\sim} N(\mu, \sigma^2) \Rightarrow g(X_n) \dot{\sim} N\left(g(\mu), |g'(\mu)|^2 \sigma^2\right)$

Mean squared error

□ $\text{MSE}(\widehat{\theta}_n) = \mathbb{E}(\widehat{\theta}_n - \theta)^2 = \text{Var}(\widehat{\theta}_n) + \text{bias}^2(\widehat{\theta}_n) = \text{Var}(\widehat{\theta}_n) + (\mathbb{E}(\widehat{\theta}_n) - \theta)^2$

Empirical Distribution Function

□ ECDF: $\widehat{F}_n(x) = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(X_i \leq x)$

□ $\mathbb{E}(\widehat{F}_n(x)) = F_X(x)$ and $\text{Var}(\widehat{F}_n(x)) = \frac{F_X(x)(1-F_X(x))}{n}$

□ Glivenko-Cantelli: $\sup_x |\widehat{F}_n(x) - F_X(x)| \xrightarrow{q.c.} 0$

□ Statistical Functional: $\theta = T(F_X)$

□ Plug-in estimator: $\widehat{\theta}_n = T(\widehat{F}_n)$

□ DKW: $\mathbb{P}\left(\sup_x |\widehat{F}_n(x) - F_X(x)| < t\right) \geq 1 - 2e^{-2nt^2}$

□ $L(x) = \max\{\widehat{F}_n(x) - t_n, 0\}$ and $U(x) = \min\{\widehat{F}_n(x) + t_n, 1\}$

$\mathbb{P}(L(x) \leq F_X(x) \leq U(x) \text{ for each } x) \geq 1 - \alpha$

where $t_n = \sqrt{\frac{1}{2n} \log\left(\frac{2}{\alpha}\right)}$.

Nonparametric Bootstrap

Real world

□ True Distribution: $F_X(\cdot)$

□ Sample: $\{X_1, \dots, X_n\} \text{ IID } F_X(\cdot)$

□ ECDF: $\widehat{F}_n(\cdot)$

□ Parameter: $\theta = T(F_X)$

□ Estimator: $\widehat{\theta}_n = T(\widehat{F}_n)$

□ Bias: $\mathbb{E}_{F_X}(\widehat{\theta}_n) - \theta$

□ Variance: $\text{Var}_{F_X}(\widehat{\theta}_n)$

Confidence Intervals

□ Normal-based Interval: $C(\alpha) = (\widehat{\theta}_n - z_{\alpha/2} \widehat{\text{se}}_{\text{BOOT}}, \widehat{\theta}_n + z_{\alpha/2} \widehat{\text{se}}_{\text{BOOT}})$

□ Pivotal Interval: $C(\alpha) = (2 \cdot \widehat{\theta}_n - \theta_{1-\alpha/2}^*, 2 \cdot \widehat{\theta}_n - \theta_{\alpha/2}^*)$

Bootstrap world

□ True Distribution: $\widehat{F}_n(\cdot)$

□ Sample: $\{X_1^*, \dots, X_n^*\} \text{ IID } \widehat{F}_n(\cdot)$

□ ECDF: $F_n^*(\cdot)$

□ Parameter: $\widehat{\theta}_n = T(\widehat{F}_n)$

□ Estimator: $\theta_n^* = T(F_n^*)$

□ Bias: $\mathbb{E}_{\widehat{F}_n}(\theta_n^*) - \widehat{\theta}_n$

□ Variance: $\text{Var}_{\widehat{F}_n}(\theta_n^*)$

□ Percentile Interval: $C(\alpha) = (\theta_{\alpha/2}^*, \theta_{1-\alpha/2}^*)$

Likelihood function and MLE

□ Likelihood: $\mathcal{L}(\theta | \mathbf{X}_n) = \prod_{i=1}^n f(X_i | \theta)$

□ MLE: $\widehat{\theta}_n = \underset{\theta}{\text{argmax}} \mathcal{L}(\theta | \mathbf{X}_n)$

□ Score function: $s(X | \theta) = \frac{\partial}{\partial \theta} \log f(X | \theta)$

□ $\widehat{\text{se}} = \left[\mathcal{I}_n(\widehat{\theta}_n)\right]^{-1/2}$

□ Asymptotic Confidence Interval: $\widehat{\theta}_n \pm z_{\alpha/2} \widehat{\text{se}}$

□ Log-likelihood: $\ell(\theta | \mathbf{X}_n) = \log \mathcal{L}(\theta | \mathbf{X}_n) = \sum_{i=1}^n \log f(X_i | \theta)$

□ Equivariance: $\widehat{\theta}_n \text{ MLE of } \theta \Rightarrow \widehat{\tau}_n = h(\widehat{\theta}_n) \text{ MLE of } \tau = h(\theta)$

□ Fisher Information: $\mathcal{I}_n(\theta) = n \mathcal{I}(\theta) = -n \mathbb{E}\left(\frac{\partial}{\partial \theta} s(X | \theta)\right)$

□ Asymptotic Normality for MLE: $\widehat{\theta}_n \dot{\sim} N(\theta, \widehat{\text{se}})$

□ If $\widehat{\tau}_n = h(\widehat{\theta}_n) \text{ MLE of } \tau = h(\theta) \Rightarrow \widehat{\tau}_n \dot{\sim} N(\tau, |h'(\widehat{\theta}_n)| \cdot \widehat{\text{se}}(\widehat{\theta}_n))$

Parametric Bootstrap

□ Sample from $f(x | \widehat{\theta}_n)$ instead from $\widehat{F}_n(\cdot)$ where $\widehat{\theta}_n$ could be the MLE.

CI's for the mean of a Normal population

Normal data, realization of a random sample $\mathbf{X}_n = \{X_1, \dots, X_n\}$ IID from $N(\mu, \sigma^2)$. Want to make inference on μ .

□ *Known variance:* $C(\alpha) = \left(\bar{X}_n - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}, \bar{X}_n + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right)$

□ *Unknown variance:* $C(\alpha) = \left(\bar{X}_n - t_{n-1, \alpha/2} \frac{S_n}{\sqrt{n}}, \bar{X}_n + t_{n-1, \alpha/2} \frac{S_n}{\sqrt{n}}\right)$

Hypothesis Testing: $H_0 : \theta \in \Theta_0$ vs $H_1 : \theta \in \Theta_1$

- Rejection region: $\mathcal{R} = \{x_n : T(x_n) > c\}$; $T(\cdot)$ test statistics, c critical value, x_n realization of random sample of size n .
 - Power function: $\beta(\theta) = \mathbb{P}_\theta(X_n \in \mathcal{R})$
 - Power of the test: $1 - \mathbb{P}(\text{Type II error}) = 1 - \beta = \inf_{\theta \in \Theta_1} \beta(\theta)$
 - Size of the test: $\mathbb{P}(\text{Type I error}) = \alpha = \sup_{\theta \in \Theta_0} \beta(\theta)$
 - p -value = $\inf\{\alpha : T(x_n) \in \mathcal{R}_\alpha\} = \sup_{\theta \in \Theta_0} \mathbb{P}_\theta(T(X_n) > T(x_n))$
 - Wald test: Reject $H_0 : \theta = \theta_0$ when $|W| > z_{\alpha/2}$ where $W = \frac{\widehat{\theta}_n - \theta_0}{\widehat{\text{se}}}$ and $\widehat{\theta}_n$ is an asymptotically Normal estimator for θ .
 - FWER: $\mathbb{P}(\text{Number of false discoveries} \geq 1) \leq \alpha$
 - Bonferroni: Given p -values $\{p_1, \dots, p_m\}$. Reject $H_{0,k}$ if $p_k < \frac{\alpha}{m}$.
 - FDR: $\mathbb{E}(\text{False discovery proportion}) \leq \alpha$
 - Benjamini Hochberg: Given ordered p -values $p^{(1)} \leq \dots \leq p^{(m)}$, let $j = \max\{k : p^{(k)} < k \cdot \frac{\alpha}{m}\}$ and $t_{\text{BH}} = p^{(j)}$. Reject $H_{0,k}$ if $p_k < t_{\text{BH}}$.
 - Likelihood Ratio Test (LRT): Reject $H_0 : \theta = \theta_0$ when $T_n > \chi_{1,1-\alpha}^2$ where $T_n = -2 \log \lambda(X_n)$ and $\lambda(X_n) = \frac{\sup_{\theta \in \Theta_0} \mathcal{L}(\theta)}{\sup_{\theta \in \Theta} \mathcal{L}(\theta)} = \frac{\mathcal{L}(\widehat{\theta}_0)}{\mathcal{L}(\widehat{\theta}_{\text{MLE}})}$.
 - Permutation Test $H_0 : F = G$. Let $\{X_i\}_{i=1}^n \sim F$, $\{Y_j\}_{j=1}^m \sim G$, $Z = \{X_1, \dots, X_n, Y_1, \dots, Y_m\}$, and $L = \{1, \dots, 1, 2, \dots, 2\}$ group labels.
Let $T = g(Z, L)$ be any test statistics and t_{obs} its value on the original data.
Get K random permutations $\{\pi_k\}_{k=1}^K$ of the labels and evaluate $T_k = g(Z, L_{\pi_k})$, then compute the p -value as: $p = \frac{1}{K} \sum_{k=1}^K \mathbb{I}(T_k > t_{\text{obs}})$.
 - χ^2 GOF: Test $H_0 : f(x | \theta_0)$ is the correct model. Let $\{I_k\}_{k=1}^K$ be K disjoint intervals, $p_k = \int_{I_k} f(x | \theta_0) dx$ and $N_k = \#\{\text{obs in } I_k\}$.
Under the null, the test statistics $Q_n = \sum_k \frac{(N_k - n \cdot p_k)^2}{n \cdot p_k} \xrightarrow{d} \chi_{K-1}^2$. Reject H_0 when $Q_n > \chi_{K-1,1-\alpha}^2$ with p -value = $\mathbb{P}(\chi_{K-1}^2 > q_{\text{obs}})$.
-